



Semester 1 Examinations 2015 / 2016

Exam Code(s)	1SD1/1SD3/1MF1/1IT1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development, Master of Information Technology
Module Code(s) Module(s)	CT874 Programming I
Paper No.	I
External Examiner(s) Internal Examiner(s)	Professor Liam Maguire Professor Gerard Lyons Dr. Michael Madden * Seamus Hill
<u>Instructions:</u>	Candidates are required to answer: Section A (20 Marks) Answer all 10 parts of section A. This section consists of 10 multiple-choice questions, all of which should be attempted. Answers are to be written on the MCQ sheet provided, NOT on the Examination paper. Section B (80 Marks) Answer Any Two Questions.
Duration No. of Pages Department(s)	2 hours 7 Information Technology
Requirements	MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

Q1.

- 1) Consider the following code:

```
public class X {  
    public void parentMethod(){  
        System.out.println("parent");  
    }  
}  
  
class Y extends X {  
    public static void main(String args[]){  
        Y y = new Y();  
        y.parentMethod();  
    }  
  
    public static void myMethod(){  
        System.out.println("child");  
    }  
}
```

What is the output of this code?

- a) parent
 - b) child
 - c) X
 - d) There is no output
- 2) How would we write the first line of a class, Graduate, such that it is a subclass of the class, Student?
- a) public Student extends Graduate{
 - b) public Student implements Graduate{
 - c) public Graduate extends Student{
 - d) public Graduate implements Student {
- 3) What is the output of the following code:

```
String a = "piano";  
  
for(int i=1; i<a.length(); i++){  
    System.out.print(a.charAt(i+1));  
}
```

- a) The output will be "piano"
- b) The output will be piano
- c) The output will be iano
- d) An exception will be thrown

- 4) What is the output of this code:

```
public class Array {  
    public static void main(String[] args){  
  
        float[] a = new float[3];  
        float x = 0, y = x;  
  
        for(int i=0; i<a.length; i++){  
            x += a[i];  
        }  
        System.out.print(y += x/a.length);  
    }  
}
```

- a) An exception will be thrown
- b) 3.0
- c) 0.0
- d) There is no output

- 5) Examine the following code:

```
class Bus extends Vehicle {  
  
    public Bus(){  
        super();  
    }  
}
```

What does super do?

- a) Makes a call to the Object() constructor.
 - b) Makes a call to the Bus() constructor.
 - c) Makes a call to the Vehicle() constructor.
 - d) Makes a call to the superclass of Vehicle's constructor.
- 6) Complete the following sentence: A(n) _____ has a method prototype but no body.
- a) default constructor
 - b) default method
 - c) concrete method
 - d) abstract method
- 7) The practice of writing multiple constructors to handle different sets of inputs is known as _____.
- a) Inheritance
 - b) Polyinputting
 - c) overloading
 - d) multiplicity

8) What is the problem with the following code:

```
public class Bag {  
    private String colour;  
  
    private Bag(){  
        colour = "unknown";  
    }  
  
    public String getColour(){  
        return colour;  
    }  
  
    class New {  
        Bag b;  
  
        public void getBag(){  
            return b;  
        }  
    }  
}
```

- a) The accessor method needs a different return value.
- b) b has not yet been initialized.
- c) The class needs more methods.
- d) The method getBag needs more statements.

9) What is the output of the following code?

```
public class TryCatch {  
  
    public static void main(String[] args){  
  
        try{  
            int number1 = Integer.parseInt("I542");  
            System.out.print("number1 is ok ");  
        }catch(NumberFormatException e){  
            System.out.print("Error");  
        }  
    }  
}
```

- a) number1 is ok
- b) There is no output
- c) Error
- d) number1 is ok Error

10) What is meant by “returning an object” from a method?

- a) It means you are returning a null value to the caller.
- b) It means returning a primitive type to the caller.
- c) It means returning the object’s address to the caller.
- d) It means returning the object’s value to the caller.

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) In relation to object-oriented programming with Java, detail your understanding, using examples/analogies where appropriate, of each of the following:
- i. Abstraction
 - ii. Polymorphism
 - iii. Inheritance
 - iv. Encapsulation

[16 Marks]

- b) Write an application that uses an array data structure to store 12 numerical values entered by the user. Each of these values represents the amount of rainfall for a particular month of the year.

The application should calculate and output the average rainfall for the twelve month period.

[10 Marks]

- c) Write a program that replies either *Leap Year* or *Not a Leap Year* when given a year. It is a leap year if the year is evenly divisible by 4 but not by 100 (i.e. 1796 is a leap year because it is divisible by 4 but not by 100). A year that is divisible by both 4 and 100 is a leap year if it is also divisible by 400 (i.e. 2000 is a leap year but 1800 is not).

[14 Marks]

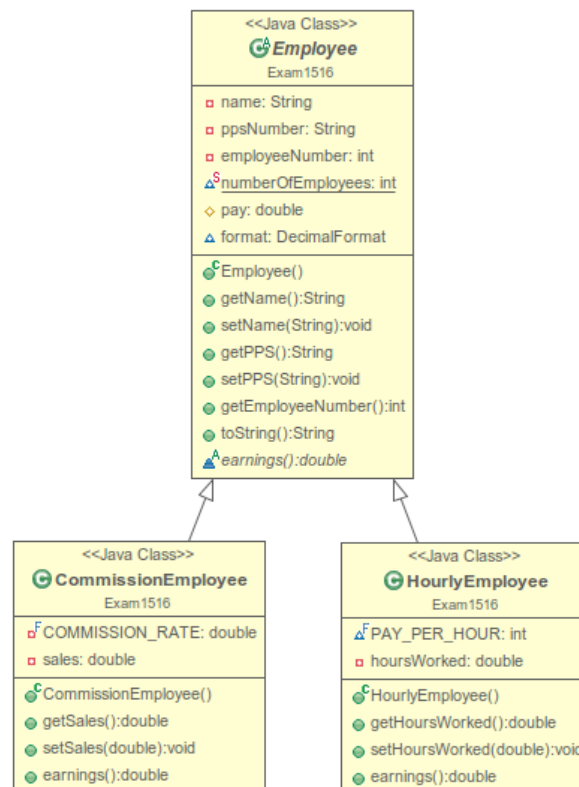
PTO

Q 3.

- a) Explain the terms **overloading** and **overriding** as used in object-oriented programming. You should provide snippets of Java code to illustrate your answer.

[10 Marks]

- b) Irish Sales Ltd. require a payroll application written in Java. The workforce is made up of two different types of employees, an hourly employee, who is paid by the hour and a commission employee, who is paid only on a commission basis. Hourly employees are paid €15.00 per hour and commission based employees are paid 10% of the value of their sales figure.



- a) You are required to implement the inheritance hierarchy illustrated above ensuring Employee implements the interface Serializable.

[12 Marks]

- b) Test your classes using a *client class* and store instances of different Employee types into a LinkedList.

[8 Marks]

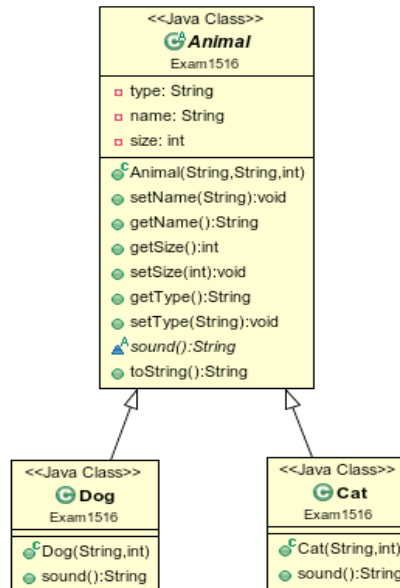
- c) Add code to allow your client class to *serialize* your application.

[10 Marks]

PTO

Q 4.

ACME pet shop is looking for a Java application to keep track of its stock of Dogs & Cats.



- a) You are required to implement the inheritance hierarchy illustrated above.

[15 Marks]

- b) Write a *tester class* which creates a LinkedList of Animal Types and contains the following methods, which should all be invoked from main():

[10 Marks]

- *addAnimals(List<Animal> list)* - populates the LinkedList with Dog & Cat objects using a while loop.

[5 Marks]

- *writeToFile(List<Animal> list)* - writes the contents of the collection to a text file.

[5 Marks]

- *List<Animal> readFromFile()* - read data from the text file and store it in a LinkedList.

[5 Marks]